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SUMMARY

Versatile and impact-driven IoT Fullstack Developer with a solid foundation in Informatics from UIN Sunan Kalijaga Yogyakarta. Demonstrates dual expertise in embedded hardware systems (ESP32, Arduino, sensor integration, IoT protocols) and modern end-to-end software architectures (React, Next.js, Node.js, Python/FastAPI, SQL Server/MySQL). Proven track record of high-caliber engineering excellence, including winning 3 national software and technology competitions. Experienced in designing robust telemetry pipelines—bridging physical sensors and microcontrollers with secure RESTful APIs, relational databases, and interactive real-time web monitoring dashboards.

CORE COMPETENCIES

- IoT & Embedded Systems: ESP32, Arduino, C/C++ for Embedded, Microcontroller Firmware Development, Sensor Integration (DHT11/DHT22, HC-SR04, DS18B20, MPU6050, RTC DS3231), Actuator & Relay Interfacing. Mobile Application Development.
- IoT Protocols & Communication: HTTP/HTTPS REST API, JSON Payload Optimization, Wi-Fi Telemetry, Periodic Data Logging, Remote Device Actuation & State Polling.
- Fullstack Web & Dashboard Engineering: React.js, Next.js (App Router), TypeScript, JavaScript, Tailwind CSS, Responsive Telemetry Dashboards, Interactive Data Charting (Chart.js), Geospatial Visualization (Leaflet/GeoJSON).
- Backend & API Systems: RESTful API Architecture, Node.js, Express.js, Python (FastAPI), Server Actions, API Proxying, Secure Device Key Management.
- Database & Data Management: SQL Server, MySQL, PostgreSQL, SQLite, Prisma ORM, Time-Series & Historical Sensor Data Logging, Relational Data Modeling.
- Mobile & Cloud Platforms: Native Android (Kotlin, Jetpack Compose), Flutter, Cloudflare R2, Supabase, Git/GitHub, Vercel CI/CD.

WORKING EXPERIENCE

Independent Fullstack Web & Mobile Developer

Jan 2024 – Present

Self-Employed / Project Engineering

- Architected and deployed 10+ production-grade systems across embedded IoT, web, and mobile platforms, spanning real-time telemetry, environmental logging, remote device control, and interactive geospatial analytics.
- Engineered end-to-end IoT pipelines connecting edge devices (ESP32/sensors) to cloud backends and React-based dashboards via secure HTTP REST APIs.
- Designed normalized relational database schemas (SQL Server, MySQL, PostgreSQL) tailored for low-latency time-series sensor logs, device status tracking, and query filtering.
- Competed in national software hackathons and engineering competitions, earning podium finishes for robust end-to-end system architectures, code reliability, and real-world societal problem-solving.
- Implemented clean Git workflows, modular codebases adhering to clean architecture, and automated deployment pipelines across Vercel and Google Play.

EDUCATION

UIN Sunan Kalijaga Yogyakarta

Aug 2022 - 2026 (Expected)

Bachelor of Science in Informatics / Computer Science (S1)

- Focus Areas: Fullstack Web Engineering, Mobile Application Development, Interactive Data Visualization, Real-Time Monitoring Systems, Algorithms & Data Structures, and Database Systems.
- Academic Activities: Active participant in national programming contests, software engineering hackathons, and developer study groups.

CERTIFICATION

- Harvard CS50x: Introduction to Computer Science — Harvard University / edX (2023)
- Certificate of Achievement: 2nd Place Winner — INTECH FEST Web Design Competition (2024)
- Certificate of Achievement: 3rd Place Winner — I/O FEST National Web Development (2025)
- Certificate of Finalist — National Innovation Week Competition (NIWC) 3.0 (2025)

TECHNICAL SKILLS

Programming Languages

TypeScript, JavaScript (ES6+), Kotlin, Dart, Python, HTML5, CSS3

Frontend Frameworks & Libraries

React, Next.js (App Router), Tailwind CSS, Chart.js, Leaflet, React-Leaflet

Mobile Engineering

Flutter, Native Android, Jetpack Compose, Material 3, Android SDK (SDK 24–36)

Backend & APIs

Node.js, Express.js, FastAPI (Python), RESTful APIs, Next.js Server Actions

Databases & Cloud Storage

PostgreSQL, SQLite, Prisma ORM, Supabase, Cloudflare R2

Tools & Methodologies

Git, GitHub, VS Code, Vite, Vercel CI/CD, Agile/Scrum

Languages

Indonesian (Native Speaker) | English (Professional Working Proficiency)

SELECTED IOT & EMBEDDED PROJECTS

ESP32 End-to-End IoT Telemetry & Monitoring Architecture

ESP32 | IoT Sensors | HTTP REST API | SQL Server | React.js

- Architected a complete closed-loop IoT ecosystem integrating multi-node ESP32 devices, a centralized HTTP REST API, SQL Server, and an interactive React web dashboard.
- Standardized bi-directional JSON payload formats for real-time sensor ingestion (POST) and dynamic web metric queries (GET).
- Built a responsive dashboard featuring multi-device status monitoring, historical time-series charts, and data filtering by device ID and timestamps.

ESP32 Multi-Sensor Environmental Data Logger

ESP32 | DHT22 / DS18B20 / MPU6050 / RTC | Wi-Fi | SQL Server | React.js

- Built an edge data logging system capturing multi-variable environmental and inertial data at scheduled intervals using an ESP32 and precision sensors.
- Optimized network reliability by bundling timestamped sensor packets with fallback queuing for uninterrupted telemetry transmission.
- Delivered a web visualizer enabling granular date-range filtering, device selection, and exportable analytical charts.

Indonesia Solar Energy Map

Interactive National Solar Irradiance Potential & Geospatial Analytics Platform

- Problem: Solar irradiance datasets across Indonesia were fragmented, complex, and inaccessible to clean energy researchers and regional project planners.
- Solution: Engineered an interactive geospatial platform mapping renewable solar potential across 37 provinces and 514 districts. Features dynamic Leaflet choropleth layers with CartoDB Dark basemaps, custom GHI gradient scales, and temporal irradiance analytics.
- Technology Used: Next.js 16+, Leaflet, React-Leaflet, Prisma ORM, TypeScript, Chart.js, GeoJSON, Open-Meteo API.
- Impact & Key Highlights:
 - Visualized Global Horizontal Irradiance (GHI), Direct Normal (DNI), and Diffuse (DHI) metrics nationwide.
 - Eliminated 38+ client request waterfalls by establishing server-side aggregation endpoints.
 - Optimized payload delivery by filtering 8.7MB boundary GeoJSON dynamically on demand.
 - Provided interactive daily, monthly, and annual radiation curve visualizers for feasibility analysis.
 - Live URL: solar-energy-map.vercel.app | Source: github.com/zakiibnu723/solar-energy-map

Lively Weather

Real-Time Global Weather Forecast & Atmospheric Dashboard

- Problem: Conventional forecast applications often expose external API keys on the client-side, suffer from third-party rate limits, and provide static, unintuitive data displays.
- Solution: Built a modern fullstack meteorological platform featuring responsive search, dynamic ambient video themes matching live weather conditions, and server-side API proxy protection with intelligent caching.
- Technology Used: Next.js 14, React 18, TypeScript, Prisma ORM, Chart.js, SQLite, Visual Crossing API.
- Impact & Key Highlights:
 - Implemented a 30-minute in-memory server caching engine with automatic fallback data to survive rate limits.
 - 100% secured API credentials by isolating third-party requests behind server routes.
 - Synchronized interactive 24-hour hourly trend visualizer with temperature and precipitation curves.
 - Comprehensive 7-day outlook including UV index, humidity, wind velocity, and atmospheric pressure.
 - Live URL: lively-weather.vercel.app | Source: github.com/zakiibnu723/Lively-Weather

Voiz.AI

Native Android Zero-Shot Voice Morphing & Audio Clone Studio

- Problem: Mobile audio manipulation tools were prone to memory crashes during audio uploads, suffered from file-handle disk locks, and lacked real-time pacing control.
- Solution: Developed a native Android application in Jetpack Compose featuring a dual-input Clone Studio (live mic recording + audio file upload), real-time pitch/tempo calibration, and low-latency Cloudflare R2 streaming.
- Technology Used: Android SDK (Kotlin), Jetpack Compose, FastAPI, Supabase, Cloudflare R2, Hugging Face.
- Impact & Key Highlights:
 - Successfully produced a signed, optimized production release bundle (VoizAI-v1.0-release.aab, 11.4MB, target SDK 36) ready for Google Play.
 - Completely resolved upload disk-locking IO exceptions by migrating from disk-streaming to memory-buffered byte requests.
 - Built an interactive tuning console allowing live tempo (0.7x–1.35x) and pitch adjustments on the fly.
 - Integrated persistent local gallery with native Android Intent sharing (WhatsApp, TikTok, Telegram).